

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DARES SALAAM REGION
FORM SIX MOCK EXAMINATION
PHYSICS 3B

Code: 131/3B

Time: 20 Hours Wednesday 21st February 2024 A.M

INSTRUCTIONS

1. This paper consist of **three (3)** questions.
2. Answer **all** questions.
3. Question number 1 carries **20 marks** and the other two (2) carry **15 marks** each.
4. Mathematical tables and non-programmable calculator may be used.
5. Cellular phones and any unauthorized material are **NOT** allowed in the examination room.
6. Write your **examination number** on every page of your answer booklet(s)
7. The following information may be useful
 - Pie $\pi = 3.14$
 - $g = 9.81 \text{ m/s}^2$

GEPAM science hub

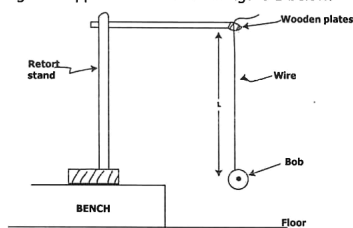
WhatsApp 0748978398

This paper consists of 3 printed pages

1. The aim of the experiment is to determine the coefficient of rigidity η of the wire provided

Proceed as follows

- (a) Arrange the apparatus as shown in figure 1 below.



- (b) Measure the length of the wire $L = 90\text{cm}$ and suspend the pendulum bob as shown in figure 1
 (c) Twist the bob in a small angular displacement so as to give the wire a shear strain and release it. Record the time $t(s)$ for 20 oscillations hence determine the periodic time $T(s)$
 (d) Repeat procedure in (c) above for $L = 80\text{cm}, 70\text{cm}, 60\text{cm}, 50\text{cm}$ and 40cm

Questions

- (i) Tabulate your results for $L(m)$, $t(s)$ and $T^2(S^2)$
 (ii) Measure and record mass M , and radius a , of the bob using the micrometer screw gauge provided.
 (iii) Plot a graph of $T^2(S^2)$ against $L(M)$ and find the shape
 (iv) Give $T^2 = \frac{8\pi IL}{\eta r^4}$, where $I = \frac{2}{5} Ma^2$, $r =$ radius of the wire. Determine the coefficient of rigidity η
 (v) State two source of errors.

2. You are provided with a beaker, liquid L, a thermometer, a stirrer and a wooden slab.

Proceed as follows:

- (a) Weight the empty beaker provided and record its mass
 (b) Pour 150ml of liquid L which is heated to about 90°C in a beaker. While stirring gently record the temperature of Liquid L after every two (2) minutes from 80°C until it falls to 50°C .
 (c) Weight the beaker with its contents and determine the mass of liquid L.
 (d) Repeat the procedure in 2 (b) and (c) above for 100ml of liquid L.

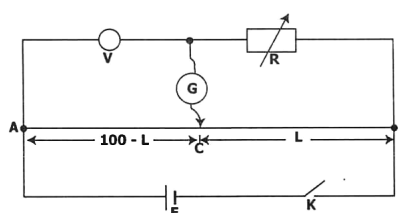
Questions.

- (i) Tabulate the results of your experiment.
 (ii) Plot cooling curve of the experiments on the same axe.
 (iii) Use the graph plotted in 2 (ii) to record the value of time taken for both volumes to fall from 80°C to 60°C as t_1 and t_2 respectively, hence compute the value of t_1/t_2 .
 (iv) Determine the ratio of mass 150ml to mass of 100ml
 (v) Compare the value obtained in 2 (iii) and 2 (iv) and give the comments.
 (vi) What is the aim of performing this experiment?

3. The aim of this experiment is to determine the internal resistance of a voltmeter.

Proceed as follows:

- (a) Connect the circuit as shown in figure 2 below



AB is the meter bridge wire, E is an accumulator, G is a Galvanometer, J is jockey and K is a switch. R is a resistance box in the right hand gap and V is a voltmeter in the left hand gap.

- (b) Start with $R = 1000\Omega$ and obtain the balances point C on the meter wire AB. Record values R , L and $100 - L$.
 (c) Repeat the experiment with $R = 2000\Omega, 3000\Omega, 4000\Omega, 5000\Omega$ and 6000Ω (voltmeter must be connected at a 3V terminal through).

Question

- (i) Tabulate your experiment results in the table shown below.

R (Ω)	1000	2000	3000	4000	5000	6000
L(CM)						
(100 - L)						
L						
100 - L						

- (ii) Plot a graph of $R(\Omega)$ against $\frac{L}{100 - L}$
 (iii) Determine the slope S of your graph.
 (iv) If R_v is the resistance of the voltmeter, obtain a formula for balance bridge in terms of balance length L and corresponding resistance R .
 (v) Use the formula in (iv) above and the graph to calculate the value R_v .
 (vi) State two source of errors and precautions take in the experiment.

PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL
GOVERNMENT DSM CITY COUNCIL
FORM VI MOCK PHYSICS 3B
(MARKING SCHEME)
2024

GEPAM science hub

WhatsApp 0748978398

1 (i) Table of result for $L(m)$, $t(s)$ and $T^2(s^2)$

L in metre	t in seconds	Period T in sec	T^2 (seconds) ²
0.9	27.28	1.36	1.86
0.8	24.91	1.25	1.55
0.7	23.25	1.16	1.35
0.6	21.97	1.10	1.21
0.5	20.22	1.01	1.02
0.4	18.15	0.91	0.82

each column takes 02 marks total = 06 marks

(ii) Mass of the bob = 70g and diameter $d = 25\text{mm}$
 So radius $a = 12.5\text{mm}$ (01 mark)

(iii) The graph is shown in graph paper behind.

From the graph

$$\text{Slope } S = \frac{\Delta T^2}{\Delta L} \quad \dots \dots \dots \text{00} \frac{1}{2}$$

$$= \frac{1.62 - 0.81}{0.84 - 0.42} = 1.93 \text{ s}^2/\text{m} \quad \dots \text{00} \frac{1}{2}$$

Range slope ($1.0 \text{ s}^2/\text{m}$ — $2.4 \text{ s}^2/\text{m}$)

(iv) Given $T^2 = \frac{8\pi I L}{\eta r^4} \quad \dots \dots \text{00} \frac{1}{2}$

Where $I =$ moment of inertia of the bob.

$$= \frac{2}{5} M a^2$$

$r =$ radius of the wire

then $\frac{T^2}{L} = \frac{8\pi I}{\eta r^4}$, but $\frac{T^2}{L} = \text{slope, "S"}$ $\dots \dots \text{00} \frac{1}{2}$

$$S = \frac{8\pi I}{\eta r^4}$$

$$\eta = \frac{8\pi I}{r^4}$$

$$\eta = \frac{8\pi}{r^4} \left(\frac{2}{5} \times 70 \times 10^3 \times (12.5 \times 10^{-3})^2 \right)$$

$$= \frac{16\pi \times 70 \times 10^3 \times (12.5 \times 10^{-3})^2}{5 \times (0.17 \times 10^{-3})^4 \times 1.93} \dots \infty \frac{1}{2}$$

$$= 68,212,623,513.8 \text{ Pa} \dots \infty \frac{1}{2}$$

$$= 0.68 \times 10^{11} \text{ Pa}$$

Range of Coefficient of rigidity η is $(0.5 \times 10^{11} - 1 \times 10^{11}) \text{ Pa}$.

(V) Sources of errors

- Parallax due to reading of metre rule
- Time reaction (starting or last ending of pressing watch)
- Air resistance

Any two @ $\infty \frac{1}{2} = 01 \text{ mark}$.

(Sec)²

THE GRAPH OF T² AGAINST L

Scales: HZ

1cm represent 0.06m

VH

1cm represent 1.09s²

(84, 162)

(42, 0.81)

T	01
IX	04
SC	04
TP	03
BL	07
ST	01
OS	

0.06 0.12 0.18 0.24 0.30 0.36 0.42 0.48 0.54 0.60 0.66 0.72 0.78 0.84 0.90 0.96 L in m

3/7

(1 cm x 1 cm) squares of (10mm x 10mm)

2) Table of result

Time in minutes	0	2	4	6	8	10	12	14	16
θ in 150 mills	80	75	70	66	63	60	56	54	51
θ in 100 mills	80	72	67	62	59	56	54	52	50

each row 03 marks = (06 marks)

(iii) From the graph $t_1 = 10$ minutes and $t_2 = 7$ minutes
then $\frac{t_1}{t_2} = \frac{10}{7} = 1.43$. . . 01 mark

Range (1 — 2).

(iv) The mass of 150 mills = 246g and for 100 mills = 198g

$$\text{The ratio } \frac{M_{150}}{M_{100}} = \frac{246}{198} \dots 00\frac{1}{2}$$

$$= 1.24 \dots 00\frac{1}{2}$$

Range (1 to 2).

(v) The values of $\frac{t_1}{t_2} = 1.43$ $\frac{M_{150 \text{ mills}}}{M_{100 \text{ mills}}} = 1.24 \dots 01 \text{ mark}$

Comments

The values in (iii) and (iv) are approximately equal
... 01 mark

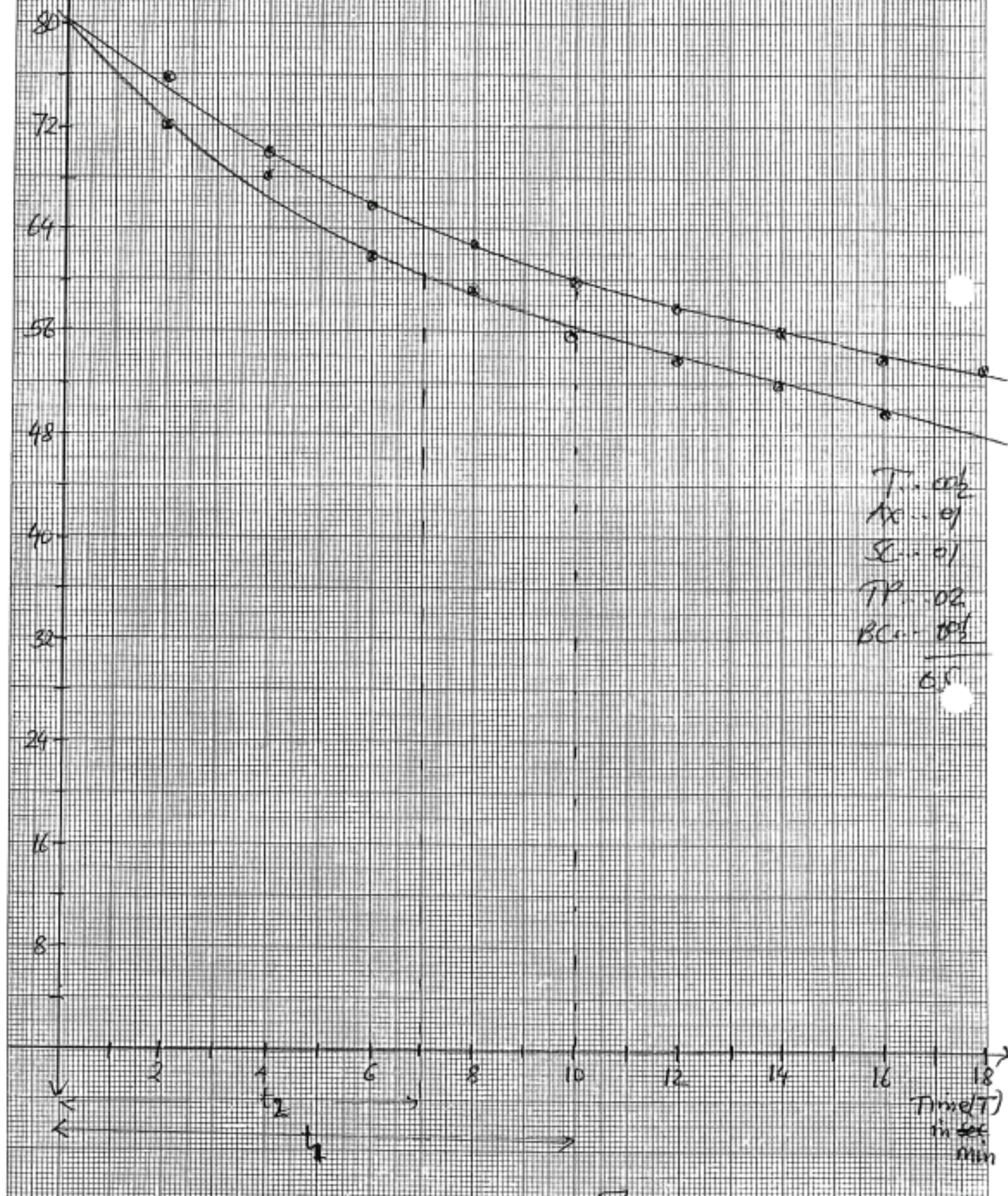
(vi) The aim of this experiment is to compare
the rate of cooling of liquid for different volumes.
... 01 mark.

0°C

THE GRAPH OF TEMPERATURE (°C) AGAINST TIME (S)

HS: 1cm represents 1 second

VS: 1cm represents 4°C



5/7

(1cm x 1cm) squares of (10mm x 10mm)

3B: 3(i) Table of result

$R (\Omega)$	1000	2000	3000	4000	5000	6000
L in cm	17.5	33	43	47	52	60
$100 - L$	82.5	67	57	53	48	40
$\frac{L}{100 - L}$	0.21	0.49	0.75	0.89	1.08	1.5

(iii) The slope $S = \frac{\Delta R}{\Delta \left(\frac{L}{100 - L} \right)}$... 00½ (05 marks)

$$S = \frac{5500 - 2500}{1.3 - 0.6} = \frac{3000}{0.7} \dots 00\frac{1}{2}$$

$$= 4285.7 \Omega \dots 00\frac{1}{2}$$

$$\approx 4300 \Omega$$

Range of slope (3000Ω - 5000Ω)

(iv) From the balancing bridge ... 00½

$$\frac{R}{L} = \frac{R_V}{100 - L}$$

$$R = R_V \left(\frac{L}{100 - L} \right) \dots 00\frac{1}{2}$$

(v) The average value of R_V stands for slopes,

So $R_V = \text{slope} \dots 00\frac{1}{2}$

$$R_V = 4300 \Omega \dots 01 \text{ mark}$$

Range R_V (3000Ω - 5000Ω).

(vi) Two sources of errors

- Parallax in reading Voltmeter.
- Zero error due to instrument

any two @ 00½ = 01 mark

THE GRAPH OF $R(\Omega)$ AGAINST $\frac{L}{100-L}$

SUBJECT NAME _____

INDEX NUMBER _____

Scale: HS: 1cm represents 0.1
VS: 1cm represents 500 Ω

